

Data Replication in a Single-Leader Environment

MongoDB Replica Set + Python Control Node

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CENG465 — Principles of Data-Intensive Systems

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Project Overview

Goal: demonstrate single-leader replication on two physical macOS machines — no containers, real LAN.

- MongoDB rs0 replica set (PRIMARY + SECONDARY)
- Schema with version, log-index, and operation-id tracking
- Python/Flask control node & live dashboard
- Write concern toggle: `w=majority` vs `w=1`
- Background reconciler for async catch-up detection

Core question

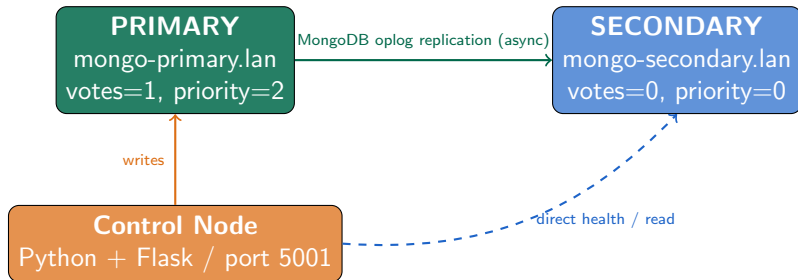
What happens when the follower goes offline?

- `w=majority`: reject the write
- `w=1`: accept the write, queue it
- How do we **observe** catch-up?

Machines

Primary	192.168.88.146
Secondary	192.168.88.105

Deployment Architecture



Why non-voting secondary?

With two voting members, disconnecting secondary removes majority — primary steps down and even $w=1$ cannot write. $votes=0$ keeps primary writable during outages.

Consequence

MongoDB alone satisfies $w=majority$ from the primary. We add an application-level guard: reject before any primary mutation if secondary cannot be pinged.

Replica Set Configuration

Initial issue

- Two voting members: primary + secondary
- Disconnecting secondary removed majority
- MongoDB reported ReplicaSetNoPrimary
- Even w=1 could not write — no writable primary

```
cfg = rs.conf()
cfg.members[1].priority = 0
cfg.members[1].votes = 0
rs.reconfig(cfg)
```

Solution: non-voting read replica

- Primary: votes=1, priority=2
- Secondary: votes=0, priority=0

Primary stays writable during follower outages.

Domain Schema — Collections & Consistency Map

Collection	Write Freq.	Consistency Need	Experiment
vehicles	Low	Eventual consistency OK	Exp. 2
drivers	Low	Sync, lookup-critical	Exp. 1
depots	Near-static	Sync, rarely changes	Exp. 1
shipments	Medium	Monotonic reads	Exp. 4
positions	High	Eventual consistency OK	Exp. 5
incidents	Low	Read-after-write	Exp. 3

Design principle

Write frequency and business criticality drive the consistency model.

Universal document fields

```
{
  key,           -- business identifier
  value,         -- domain payload
  version,       -- increments each write
  leader_term,   -- Raft epoch
  last_log_index, -- write ordering
  last_operation_id,
  last_updated,
  deleted        -- soft delete
}
```

operation_logs (observability)

```
{
  operation_type, -- insert/update/delete
  log_index,
  leader_write_time,
  follower_visible_time,
  replication_delay_ms,
  status, -- pending_follower
           -- visible_on_follower
           -- timeout
  write_concern -- majority / 1
}
```

Write Concern Modes

Mode	Secondary Up	Secondary Down
w=majority	Write succeeds; replication delay is measured and logged	Write rejected before primary mutation — no data change
w=1	Write succeeds immediately; log is reconciled in background	Write succeeds on primary; log stays pending_follower until catch-up

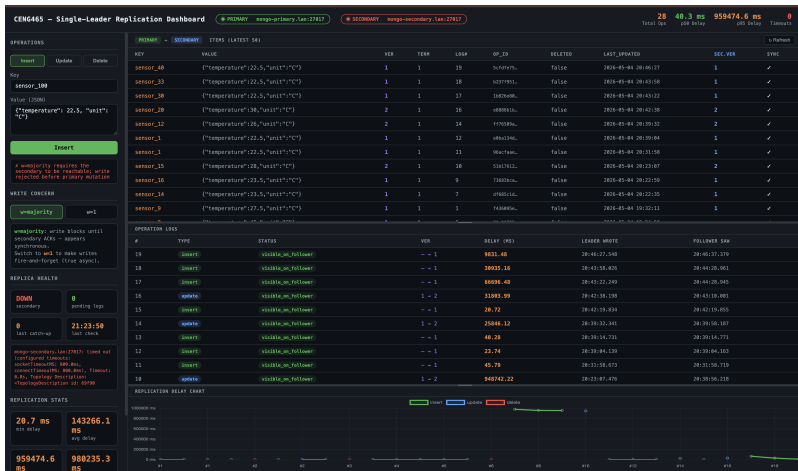
w=majority — Consistency-first

Reject the write rather than risk divergence. No partial state.

w=1 — Availability-first

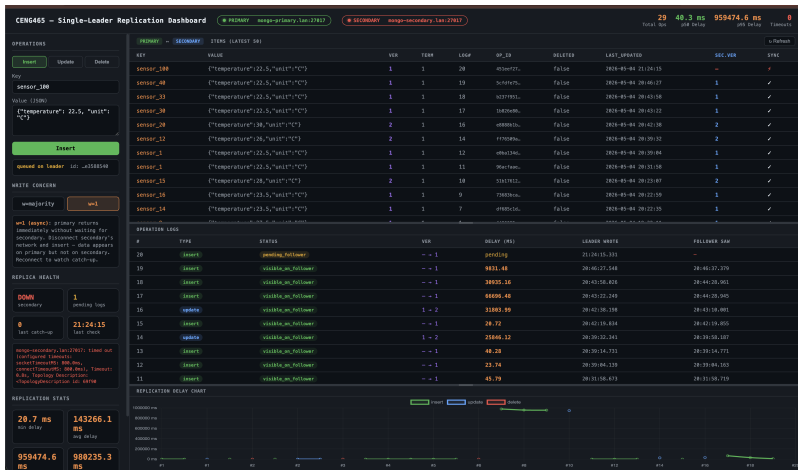
Accept the write; guarantee eventual delivery via oplog.

Demo Step 1 — w=majority Write Rejected



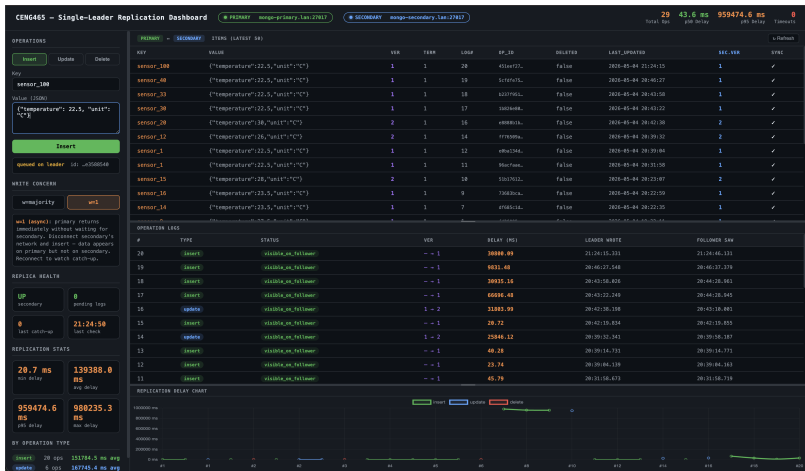
Secondary offline. Insert **rejected before primary mutation** — no new item, no pending log created.

Demo Step 2 — w=1 Write Succeeds



Switched to w=1. Insert completes instantly: sensor_100 on primary (ver=1), sec.ver = ---. Log: pending_follower.

Demo Step 3 — Secondary Reconnects & Catches Up



Oplog replayed; reconciler detects version match. Log: visible_on_follower; sec.ver matches; sync ✓ restored.

What we proved

- Replication is **asynchronous by default**
- Write concern controls the **durability–availability tradeoff**
- $w=\text{majority}$: consistency over availability
- $w=1$: availability with eventual catch-up via oplog
- **Observability** (version, delay, log-index) is essential

Implementation highlights

- Application-level write guard for $w=\text{majority}$
- Background reconciler (1 s cadence)
- `directConnection=true` for honest secondary health
- Version + log-index fallbacks for catch-up detection

Key insight

Single-leader replication gives strong write ordering but does **not** guarantee read-after-write consistency on followers.